

Hyeonseok Moon

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Research Area

- Language Resource
- Benchmark
- Large Language Model
- Language Generation
- Model Evaluation
- Data Evaluation
- Machine Translation
- Data Engineering

Work Experience

Samsung Electronics - Mobile eXperience Business 2026.03 – now
Staff Engineer. Language AI Team
- AI Integration & Workflow Automation - Bixby Development - Data Analysis & Error Detection

Education

Korea University (Seoul, Republic of Korea) 2021.03 – 2026.02
Ph.D.
Major in Computer Science and Engineering - Advisor: Prof. Heuseok Lim

Korea University (Seoul, Republic of Korea) 2015.03 – 2021.02
Bachelor of Science and Engineering
Major in Mathematics and Artificial Intelligence (Double Majors) - Advisor: Prof. Euisung Park

Selected Publications

International Conference

- NeedleChain: Measuring Intact Context Comprehension Capability of Large Language Models** 2026
Hyeonseok Moon, Heuseok Lim
ACL 2026 Findings
- Towards Privacy-Preserving Large Language Model: Text-free Inference Through Alignment and Adaptation** 2026
JeongHo Yoon, Chanhee Park, Yongchan Chun, Hyeonseok Moon (+), Heuseok Lim(+)
ACL 2026
- Beyond Hard Negatives: The Importance of Score Distribution in Knowledge Distillation for Dense Retrieval** 2026
Youngjoon Jang, Seongtae Hong, Hyeonseok Moon(+), Heuseok Lim(+)
SIGIR 2026 Short
- Improving Semantic Proximity in Information Retrieval through Cross-Lingual Alignment** 2026
Seongtae Hong, Youngjoon Jang, Jungseob Lee, Hyeonseok Moon(+), Heuseok Lim(+)
ICLR 2026
- Metric Calculating Benchmark: Code-Verifiable Complicate Instruction Following Benchmark for Large Language Models** 2025
Hyeonseok Moon, Seongtae Hong, Jaehyung Seo, Heuseok Lim
EMNLP 2025
- LimaCost: Data Valuation for Instruction Tuning of Large Language Models** 2025
Hyeonseok Moon, Jaehyung Seo, Seonmin Koo, Jinsung Kim, Young-kyoung Ham, jiwon moon, Heuseok Lim
EMNLP 2025 Findings
- The Impact of Negated Text on Hallucination with Large Language Models** 2025
Jaehyung Seo, Hyeonseok Moon, Heuseok Lim
EMNLP 2025
- Call for Rigor in Reporting Quality of Instruction Tuning Data** 2025
Hyeonseok Moon, Jaehyung Seo, Heuseok Lim
ACL 2025

Cross-Lingual Optimization for Language Transfer in Large Language Models Jungseob Lee, Seongtae Hong, <u>Hyeonseok Moon</u> , Heuiseok Lim <i>ACL 2025</i>	2025
Semantic Aware Linear Transfer by Recycling Pre-trained Language Models for Cross-lingual Transfer Seungyoon Lee, Seongtae Hong, <u>Hyeonseok Moon</u> , Heuiseok Lim <i>ACL 2025 Findings</i>	2025
Find the Intention of Instruction: Comprehensive Evaluation of Instruction Understanding for Large Language Models <u>Hyeonseok Moon</u> , Jaehyung Seo, Seungyoon Lee, Chanjun Park, Heuiseok Lim <i>NAACL 2025 Findings</i>	2025
FLEX: A Benchmark for Evaluating Robustness of Fairness in Large Language Models Dahyun Jung, Seungyoon Lee, <u>Hyeonseok Moon</u> , Chanjun Park, Heuiseok Lim <i>NAACL 2025 Findings</i>	2025
MIRAGE: A Metric-Intensive Benchmark for Retrieval-Augmented Generation Evaluation Chanhee Park, <u>Hyeonseok Moon</u> , Chanjun Park, Heuiseok Lim <i>NAACL 2025 Findings</i>	2025
MIGRATE: Cross-Lingual Adaptation of Domain-Specific LLMs through Code-Switching and Embedding Transfer Seongtae Hong, Seungyoon Lee, <u>Hyeonseok Moon</u> , Heuiseok Lim <i>COLING 2025</i>	2025
Translation of Multifaceted Data without Re-Training of Machine Translation Systems <u>Hyeonseok Moon</u> , Seungjun Lee, Seongtae Hong, Seungjum Lee, Chanjun Park, Heuiseok Lim <i>EMNLP 2024 Findings</i>	2024
Length-aware Byte Pair Encoding for Mitigating Over-segmentation in Korean Machine Translation Jungseob Lee, <u>Hyeonseok Moon</u> (<i>equal contribution</i>), Seungjun Lee, Chanjun Park, Sugyeong Eo, Hyunwoong Ko, Jaehyung Seo, Seungyoon Lee, Heuiseok Lim <i>ACL 2024 Findings</i>	2024
Generative Interpretation: Toward Human-Like Evaluation for Educational Question-Answer Pair Generation <u>Hyeonseok Moon</u> , Jaewook Lee, Sugyeong Eo, Chanjun Park, Jaehyung Seo, Heui-Seok Lim <i>EACL 2024 Findings</i>	2024
Hyper-BTS Dataset: Scalability and Enhanced Analysis of Back Transcription (BTS) for ASR Post-Processing Chanjun Park, Jaehyung Seo, Seolhwa Lee, Junyoung Son, <u>Hyeonseok Moon</u> , Sugyeong Eo, Chanhee Lee, Heui-Seok Lim <i>EACL 2024 Findings</i>	2024
Leveraging Pre-existing Resources for Data-Efficient Counter-Narrative Generation in Korean Seungyoon Lee, Chanjun Park, DaHyun Jung, <u>Hyeonseok Moon</u> , Jaehyung Seo, Sugyeong Eo, Heui-Seok Lim <i>LREC-COLING 2024</i>	2024
Detecting Critical Errors Considering Cross-Cultural Factors in English-Korean Translation Sugyeong Eo, Jungwoo Lim, Chanjun Park, Dahyun Jung, Seonmin Koo, <u>Hyeonseok Moon</u> , Jaehyung Seo, Heui-Seok Lim <i>LREC-COLING 2024</i>	2024
CHEF in the Language Kitchen: A Generative Data Augmentation Leveraging Korean Morpheme Ingredients Jaehyung Seo, <u>Hyeonseok Moon</u> , Jaewook Lee, Sugyeong Eo, Chanjun Park, Heui-Seok Lim <i>EMNLP 2023</i>	2023

KEBAP: Korean Error Explainable Benchmark Dataset for ASR and Post-processing Seonmin Koo, Chanjun Park, Jinsung Kim, Jaehyung Seo, Sugyeong Eo, <u>Hyeonseok Moon</u> , Heui-Seok Lim <i>EMNLP 2023</i>	2023
Post-hoc Utterance Refining Method by Entity Mining for Faithful Knowledge Grounded Conversations Yoonna Jang, Suhylene Son, Jeongwoo Lee, Junyoung Son, Yuna Hur, Jungwoo Lim, <u>Hyeonseok Moon</u> , Kisu Yang, Heuiseok Lim <i>EMNLP 2023</i>	2023
PEEP-talk: A situational dialogue-based chatbot for English education Seungjun Lee, Yoonna Jang, Chanjun Park, Jungseob Lee, Jaehyung Seo, <u>Hyeonseok Moon</u> , Sugyeong Eo, Seounghoon Lee, Bernardo Yahya, Heui-Seok Lim <i>ACL 2023 - Demo</i>	2023
Towards diverse and effective question-answer pair generation from children storybooks Sugyeong Eo, <u>Hyeonseok Moon</u> (<i>equal contribution</i>), Jinsung Kim, Yuna Hur, Jeongwook Kim, Songeun Lee, <u>Changwoo Chun</u> , Sungsoo Park, Heuiseok Lim <i>ACL 2023 Findings</i>	2023
Improving Formality-Sensitive Machine Translation Using Data-Centric Approaches and Prompt Engineering Seungjun Lee, <u>Hyeonseok Moon</u> , Chanjun Park, Heuiseok Lim <i>IWSLT 2023</i>	2023
QUAK: A synthetic quality estimation dataset for korean-english neural machine translation Sugyeong Eo, Chanjun Park, <u>Hyeonseok Moon</u> , Jaehyung Seo, Gyeongmin Kim, Jungseob Lee, Heuiseok Lim <i>COILING 2022</i>	2022
A dog is passing over the jet? a text-generation dataset for korean commonsense reasoning and evaluation Jaehyung Seo, Seounghoon Lee, Chanjun Park, Yoonna Jang, <u>Hyeonseok Moon</u> , Sugyeong Eo, Seonmin Koo, Heuiseok Lim <i>NAACL 2022 Findings</i>	2022
KU X upstage's submission for the WMT22 quality estimation: Critical error detection shared task Sugyeong Eo, Chanjun Park, <u>Hyeonseok Moon</u> , Jaehyung Seo, HeuiSeok Lim <i>WMT 2022</i>	2022
Priming ancient Korean neural machine translation Chanjun Park, Seolhwa Lee, Jaehyung Seo, <u>Hyeonseok Moon</u> , Sugyeong Eo, Heui-Seok Lim <i>LREC 2022</i>	2022
Empirical Analysis of Noising Scheme based Synthetic Data Generation for Automatic Post-editing <u>Hyeonseok Moon</u> , Chanjun Park, Seolhwa Lee, Jaehyung Seo, Jungseob Lee, Sugyeong Eo, HeuiSeok Lim <i>LREC 2022</i>	2022
A Self-Supervised Automatic Post-Editing Data Generation Tool <u>Hyeonseok Moon</u> , Chanjun Park, Sugyeong Eo, Jaehyung Seo, SeungJun Lee, Heuiseok Lim <i>ICML 2022 - DataPerf Workshop</i>	2022
BTS: Back TranScription for Speech-to-Text Post-Processor using Text-to-Speech-to-Text Chanjun Park, Jaehyung Seo, Seolhwa Lee, Chanhee Lee, <u>Hyeonseok Moon</u> , Sugyeong Eo, Heuiseok Lim <i>WAT2021 - ACL Workshop</i>	2021
Should we find another model?: Improving neural machine translation performance with one-piece tokenization method without model modification Chanjun Park, Sugyeong Eo, <u>Hyeonseok Moon</u> , HeuiSeok Lim <i>NAACL 2021 - industry track</i>	2021
International Journal	
SERA: Self-referential assessment framework for bidirectional generative commonsense reasoning Jaehyung Seo, <u>Hyeonseok Moon</u> , Yoonna Jang, Heuiseok Lim <i>Knowledge-Based Systems</i>	2026

Exploiting Hanja-based Resources in Processing Korean Historic Documents written by Common Literati	2024
<u>Hyeonseok Moon</u> , Myunghoon Kang, Jaehyung Seo, Sugyeong Eo, Chanjun Park, Yeongwook Yang, Heuiseok Lim <i>IEEE Access</i>	
Doubts on the reliability of parallel corpus filtering	2023
<u>Hyeonseok Moon</u> , Chanjun Park, Seonmin Koo, Jungseob Lee, Seungjun Lee, Jaehyung Seo, Sugyeong Eo, Yoonna Jang, Hyunjoong Kim, Hyoung-gyu Lee, Heuiseok Lim <i>Expert Systems with Applications</i>	
Uncovering the Risks and Drawbacks Associated with the Use of Synthetic Data for Grammatical Error Correction	2023
Seonmin Koo, Chanjun Park, Seolhwa Lee, Jaehyung Seo, Sugyeong Eo, <u>Hyeonseok Moon</u> , Heuiseok Lim <i>IEEE Access</i>	
A survey on evaluation metrics for machine translation	2023
Seungjun Lee, Jungseob Lee, <u>Hyeonseok Moon</u> , Chanjun Park, Jaehyung Seo, Sugyeong Eo, Seonmin Koo, Heuiseok Lim <i>Mathematics</i>	
PU-GEN: Enhancing generative commonsense reasoning for language models with human-centered knowledge	2022
Jaehyung Seo, Dongsuk Oh, Sugyeong Eo, Chanjun Park, Kisu Yang, <u>Hyeonseok Moon</u> , Kinam Park, Heuiseok Lim <i>Knowledge-Based Systems</i>	
An automatic post editing with efficient and simple data generation method	2022
<u>Hyeonseok Moon</u> , Chanjun Park, Jaehyung Seo, Sugyeong Eo, Heuiseok Lim <i>IEEE Access</i>	
AI for patents: A novel yet effective and efficient framework for patent analysis	2022
Junyoung Son, <u>Hyeonseok Moon</u> , Jeongwoo Lee, Seolhwa Lee, Chanjun Park, Wonkyung Jung, Heuiseok Lim <i>IEEE Access</i>	
Plain template insertion: korean-prompt-based engineering for few-shot learners	2022
Jaehyung Seo, <u>Hyeonseok Moon</u> , Chanhee Lee, Sugyeong Eo, Chanjun Park, Jihoon Kim, Changwoo Chun, Heuiseok Lim <i>IEEE Access</i>	
K-nct: Korean neural grammatical error correction gold-standard test set using novel error type classification criteria	2022
Seonmin Koo, Chanjun Park, Jaehyung Seo, Seungjun Lee, <u>Hyeonseok Moon</u> , Jungseob Lee, Heuiseok Lim <i>IEEE Access</i>	
Mimicking infants' bilingual language acquisition for domain specialized neural machine translation	2022
Chanjun Park, Woo-Young Go, Sugyeong Eo, <u>Hyeonseok Moon</u> , Seolhwa Lee, Heuiseok Lim <i>IEEE Access</i>	
Word-level quality estimation for Korean-English neural machine translation	2022
Sugyeong Eo, Chanjun Park, <u>Hyeonseok Moon</u> , Jaehyung Seo, Heuiseok Lim <i>IEEE Access</i>	
An empirical study on automatic post editing for neural machine translation	2021
<u>Hyeonseok Moon</u> , Chanjun Park, Sugyeong Eo, Jaehyung Seo, Heuiseok Lim <i>IEEE Access</i>	
Return on Advertising Spend Prediction with Task Decomposition-Based LSTM Model	2021
<u>Hyeonseok Moon</u> , Taemin Lee, Jaehyung Seo, Chanjun Park, Sugyeong Eo, Imatitikua D Aiyanyo, Jeongbae Park, Aram So, Kyoungwha Ok, Kinam Park <i>Mathematics</i>	
Comparative analysis of current approaches to quality estimation for neural machine translation	2021
Sugyeong Eo, Chanjun Park, <u>Hyeonseok Moon</u> , Jaehyung Seo, Heuiseok Lim <i>Applied Sciences</i>	

Collaborative Project

World Best LLM - with NC AI

2025.07 – 2026.02

Supported by **National Research Foundation** - Team Leader at Korea University

- Supervise LLM training process and Evaluating trained LLMs
- Evaluation data curation. Evaluation dashboard management
- Training data curation and management

LLM Assistant for Teaching Human Consultant

2024.07 – 2025.08

Supported by **Creative Digital Lab** - Project Manager at Korea University

- Training Large Language Models with Data Curation
- Data augmentation with a few human-annotated labels

Legal Domain Vertical LLM

2024.07 – 2025.07

Supported by **KT** - Project Manager at Korea University

- Training Large Language Models with Data Curation
- Data quality check for building domain specialized LLM
- Constructing data processing pipeline

NLP for Ancient Korean Common Literati Document

2022.06 – 2024.07

Supported by **National Research Foundation** - Project Manager at Korea University

- Named entity recognition and document analysis for ancient Korean documents
- Engaged in data construction process and setup annotation standard
- Related Publication: Exploiting Hanja-Based Resources in Processing Korean Historic Documents Written by Common Literati (IEEE Access)

Persona-based Dialogue with k-Nearest-Neighbor Approach

2023.05 – 2023.12

Supported by **NC Soft** - Head Technician at Korea University

- Research on the applicability of k-nearest neighbor approach in persona dialogue
- k-nearest neighbor, persona dialogue, language generation

Domain Specialized Parallel Corpus Construction for Machine Translation

2022.06 – 2023.11

Supported by **NIA (with Minigate Corporation)** - Project Manager at Korea University

- Data evaluation and supervision in curation process
- Engaged in data construction process and setup annotation standard

Automated Question-Answer pair Data Generation System

2022.03 – 2023.02

Supported by **Hyundai Mortors** - Head Technician at Korea University

- Automated question-Answer pair generation framework, especially tailored to the educational purpose
- QA generation, Education domain
- Related Publication: Towards Diverse and Effective Question-Answer Pair Generation from Children Storybooks (ACL 2023 - findings)

User Query based Recommendation System

2022.03 – 2023.01

Supported by **FLES corporation** - Head Technician at Korea University

- Commercial item recommendation systems based on the user preference
- Recommendation system, Information retrieval

Fortune Telling Generation AI Project

2022.03 – 2023.01

Supported by **FLES corporation** - Project Manager at Korea University

- Fortune telling AI module. Encoder-Decoder generator along with LLM based generator system
- Language generation, Decoding strategy, Large language models
- Related Publication: SaJuTeller: Conditional Generation Deep-Learning based Fortune Telling Model (HCLT 2022)

Parallel Corpus Filtering and Mining Research Project

2021.12 – 2022.07

Supported by **Naver Papago** - Head Technician at Korea University

- Analysis on parallel corpus filtering methods targeting Korean-English machine translation
- Parallel corpus filtering, Machine translation
- Related Publication: Doubts on the reliability of parallel corpus filtering (Expert Systems with Applications)

Korean-Prompt-based Engineering for Few-shot Research Project

2022.05 – 2022.07

Supported by **Hyundai Motors** - Researcher at Korea University

- Few-shot prompting strategy for enhancing Korean understanding task performance
- Prompt engineering, Few shot, Language understanding
- Related Publication: <https://ieeexplore.ieee.org/abstract/document/9913979>
Plain Template Insertion: Korean-Prompt-Based Engineering for Few-Shot Learners (IEEE Access)

Information Retrieval system for Industrial Frequently Asked Question

2021.07 – 2022.03

Supported by **Data Voucher (O2O corporation)** - Researcher at Korea University

- Information retrieval system for frequently-asked QA systems
- Keyword Extraction, Information Extraction, Question Answering module

Return on Advertising Spend (ROAS) Prediction Project

2021.07 – 2022.03

Supported by **BizSpring corporation** - Head Technician at Korea University

- Regression module for return-on-advertising-spend prediction
- Keyword Extraction, Return on Advertising Spend, Regression Model
- Related Publication: <https://www.mdpi.com/2227-7390/10/10/1637>
Return on Advertising Spend Prediction with Task Decomposition-Based LSTM Model (Mathematics)

Patent document processing Research Project

2021.06 – 2021.09

Supported by **LG Innotek** - Head Technician at Korea University

- Sentence extraction and key phrase extraction module for patent documents
- Automatic Summarization, Sentence classification, Information Extraction
- Related Publication: <https://ieeexplore.ieee.org/abstract/document/9779775>
AI for Patents: A Novel Yet Effective and Efficient Framework for Patent Analysis (IEEE Access)

Teaching

Teaching Assistant at Korea University

2023.09 – 2024.06

- (DFE610-00) NLP for digital finance engineering
- (BDC101-00) Introduction to Natural Language Processing In Big Data
- (COSE461-02) Natural Language Processing

Honors & Awards

Outstanding Reviewer Award

2026

- Annual Conference of the European Chapter of the Association for Computational Linguistics

Invited Talk: Research Institute of Korean Studies(RIKS), Korea University

2026

- Natural Language Evaluation for Large Language Models

Best Paper Award

2024

- The 36th Annual Conference on Human & Cognitive Language Technology (HCLT2024)

Best Paper Award

2023

- The 35th Annual Conference on Human & Cognitive Language Technology (HCLT2023)

1st place in WMT 2022 QE Task 3, 2022

2022

- Seventh Conference on Machine Translation (WMT22) Quality Estimation Shared Task

Best Paper Award

2022

- The 34th Annual Conference on Human & Cognitive Language Technology (HCLT2022)

Best Paper Award

2021

- The 33rd Annual Conference on Human & Cognitive Language Technology (HCLT2021)

Patent

Device and Method For Generation of Diverse Question-Answer Pair

- U.S. Patent Application No. 18/585,166

Device And Method For Assessment Of Educational Question-Answering

- South Korea Patent Granted No. 10-2946098
- South Korea Patent Application No. 10-2023-0162875

Device and Method for Generating Fortune Telling Model Based n Conditional Generation Deep-Learning

- South Korea Patent Granted No. 10-2790031
- South Korea Patent Application No. 10-2022-0158977

Task Decomposition Method based Prediction of Return on Advertising Spend and Device Performing the same

- South Korea Patent Granted No. 10-2790027
- South Korea Patent Application No. 10-2021-0156657

Device and Method for Parallel Corpus Filtering Based On Semantic Similarity

- South Korea Patent Granted No. 10-2593448
- South Korea Patent Application No. 10-2022-0151593

Device and Method for Generating of Training Data for Quality Estimation In Machine Translation

- South Korea Patent Granted No. 10-2593447
- South Korea Patent Application No. 10-2021-0156657

Device and Method for Automatic Evaluation of Call Center Agent Conversation

- South Korea Patent Application No. 10-2025-0100212

Performance Evaluation Method For Large Language Models Based On Intention Catching Capability

- South Korea Patent Application No. 10-2024-0169352

Diverse and Effective Question-Answer Pair Generation System for Education

- South Korea Patent Application No. 10-2023-0024355

Device and Method for Generating Training Data For Post Editing

- South Korea Patent Application No. 10-2021-0118924